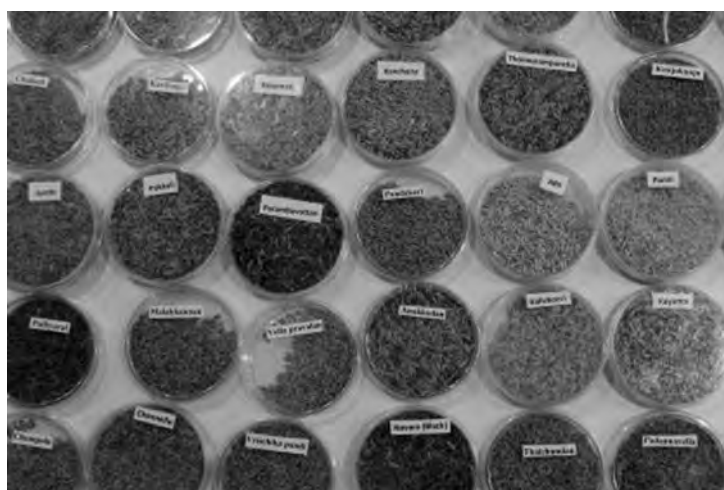


- 6 One of the aims of seed banks is to maintain genetic diversity.

Seeds are stored in the Navdanya Seed Vault, one of over a hundred seed banks in India. This seed bank stores over 4000 varieties of rice.

The photograph shows some of these varieties of rice.



- (a) (i) Which of the following can be used to calculate genetic diversity?

(1)

- ☐ A Hardy–Weinberg equation
- ☐ B heterozygosity index
- ☐ C index of diversity
- ☐ D mitotic index

- (ii) Give **two** differences between genetic diversity and species richness.

(2)

1

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2

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(b) Suggest **two** reasons why seed banks store seeds instead of growing whole plants.

(2)

1

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2

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(c) Bananas are a popular food and seed banks are involved in the development of disease-resistant varieties.

The table shows the mass of bananas imported by three countries in 2007 and in 2011.

Country	Mass imported in 2007 / tonnes $\times 10^5$	Mass imported in 2011 / tonnes $\times 10^5$
China	3.3	8.2
Iran	6.3	6.2
Japan	9.8	1.1

(i) Calculate the percentage increase in the mass of bananas imported by China.

(2)

Answer%

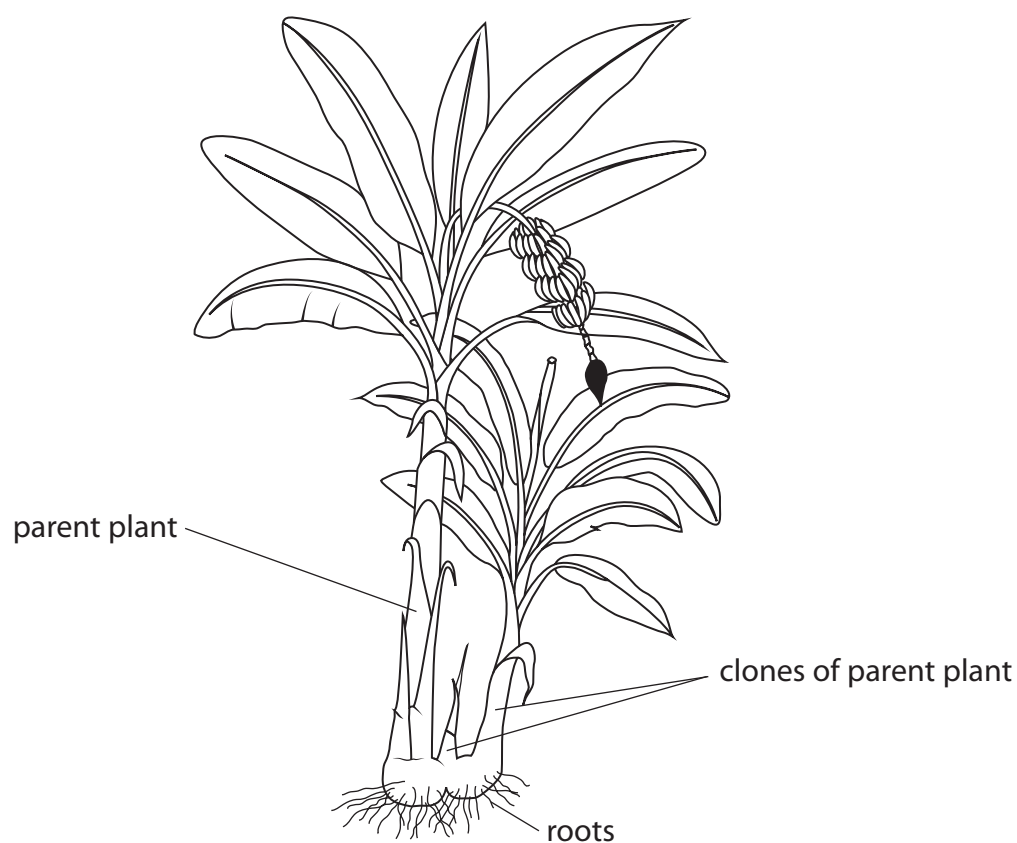


P 6 1 4 7 0 A 0 1 9 3 2

(ii) Banana plants can reproduce asexually.

Most of the bananas grown by farmers around the world are clones from a single parent banana plant called 'The Cavendish'.

The diagram shows a parent banana plant and two clones of this parent plant.



These clones can be removed from the parent plant and grown into new banana plants.

Panama disease is caused by a fungus. It has destroyed Cavendish banana plants in the Philippines. Cavendish banana plants are not resistant to this fungus.

Seed banks around the world have seeds from different varieties of banana.

Scientists are trying to improve the Cavendish banana by breeding it with different varieties of banana.



Explain why scientists are breeding the Cavendish banana with different varieties of banana.

(4)

(Total for Question 6 = 11 marks)

